DS 542 - Spring 2026 - Homework 1

Feel free to write your answers by hand. No need for LaTeX; just make sure it is legible.

1. Equation 2.5 in the textbook (Understanding Deep Learning) gives the equation for linear regression with one input variable using the least-squares loss.

$$
\begin{aligned}
L[\phi] &= \sum_{i=1}^{I} \left(f[x_i, \phi] - y_i\right)^2 \\
&= \sum_{i=1}^{I} \left(\phi_0 + \phi_1 x_i - y_i\right)^2
\end{aligned}
$$

   (a) **(2 points)** Derive the formula for $\partial L / \partial \phi_0$.

$$\frac{\partial L}{\partial \phi_0} = 2\sum_{i=1}^{I} (\phi_0 + \phi_1 x_i - y_i)$$

   (b) **(2 points)** Derive the formula for $\partial L / \partial \phi_1$.

$$\frac{\partial L}{\partial \phi_1} = 2\sum_{i=1}^{I} x_i (\phi_0 + \phi_1 x_i - y_i)$$

2. Answer the following questions about the following equation.

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 8 & 7 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} x_{0,0} & x_{0,1} \\ x_{1,0} & x_{1,1} \end{bmatrix}$$

(a) **(2 points)** What is the mathematical operation on the lefthand side of the equation?

matrix multiplication ※

(b) **(2 points)** Write out the formula to calculate $x_{1,0}$.

$x_{1,0} = 4 \times 8 + 3 \times 5 = 47$ ※

3. **(2 points)** Answer the following questions about ordinary least squares linear regression, where y_i is the target value and $\hat{y}_i$ is the predicted value. What is the equation for the mean residual on the training data?

residual $= y_i - \hat{y}_i \Rightarrow$ mean residual $= \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)$ ※

4. Answer the following questions about basic probability.

(a) **(2 points)** Suppose A and B are events with $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. Compute $P(A \mid B)$ and $P(A \cup B)$.

$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.5} = 0.4$ ※

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= 0.4 + 0.5 - 0.2 = 0.7$ ※

(b) **(2 points)** Let X be a Bernoulli random variable with $P(X = 1) = p$. What are $\mathbb{E}[X]$ and $\mathrm{Var}(X)$?

$P(X=1) = p \Rightarrow P(X=0) = 1-p$

$E(X) = 1 \cdot p + 0 \cdot (1-p) = p$ ※

$Var(X) = E(X^2) - E(X)^2 = 1^2 \cdot p + 0^2 \cdot (1-p) - p^2 = p(1-p)$ ※